

Chemical kinetics

- Chemical kinetics may also be called reaction kinetics or simply "kinetics". The rate of a chemical reaction usually has units of sec^{-1}
- In 1864, Peter Waage and Cato Guldberg pioneered the development of chemical kinetics by formulating the law of mass action, which states that the speed of a chemical reaction is proportional to the quantity of the reacting substances.
- The reaction rate is the rate at which the reactants of a chemical reaction form the products.
- Reaction rates are expressed as concentration per unit time.

Order &Molecularity

Molecularity	Order
• Molecularity of a reaction is the number of atoms, ions or molecules taking part in a reaction as in chemical equation.	• Order of reaction is the total power of concentration of reactants, which is proportional to the experimental facts.
• Molecularity is a theoretical concept which depends on chemical equation partly.	• Order of reaction is an experimental fact which depends on experiment, observation and conclusion.
• Molecularity can be fraction	• Order of reaction never be fraction
• Example: $2 \text{NH}_3(\text{g}) \rightarrow 3 \text{H}_2(\text{g}) + \text{N}_2(\text{g})$ Molecularity=2	• Example: $2 \text{NH}_3(\text{g}) \rightarrow 3 \text{H}_2(\text{g}) + \text{N}_2(\text{g})$ Order- Zero order reaction
• <u>Zero order Reaction:</u>	

The reaction whose rate is independent of the concentration of a reactant, so that changing its concentration has no effect on the speed of the reaction is called zero-order reactions.

- Example:



- **First order Reaction:**

If a reaction rate depends on a single reactant and the value of the exponent is one, then the reaction is said to be first order.

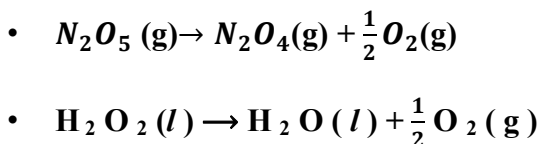
Definition: The reaction whose rate is proportional to first power of concentration of one reactant is called the first order reaction.

- According to definition :

rate of reaction:

$$R = \frac{-dc}{dt} = K \times C^1 = K \times C$$

- Example



- **Second order Reaction:**

- A reaction is said to be second order when the overall order is two. The rate of a second-order reaction may be proportional to one concentration squared to the product of two concentrations.

- **Definition:** The reaction whose rate is proportional to second power of concentration of one reactant or two reactants is called second order reaction.

According to definition,

rate of reaction $r = \frac{-dc}{dt} = k \times C^2$

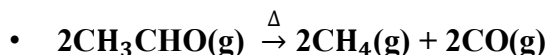
There are two types of second order reactions which are generally expressed as,

(i) $2A(\text{reactant}) \rightarrow \text{products}$

(ii) $A + B \rightarrow \text{products}$

- Examples of a Second order reaction
- $2NO_2(g) \rightarrow 2NO(g) + O_2(g)$

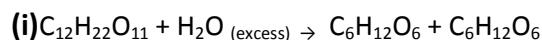
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Pseudo unimolecular reaction

Definition: It is a reaction which seems to be second degree but actually a first order reaction. One of the reactant is in excess which makes its importance negligible and to follow the first order rate is called, "Pseudo Unimolecular Reaction".

Example:



$$\text{Rate} = K [\text{C}_{12}\text{H}_{22}\text{O}_{11}]$$



$$\text{Rate} = K [\text{CH}_3\text{COOC}_2\text{H}_5]$$

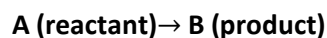
Rate of reaction: The reaction rate or speed of reaction for a reactant or product in a particular reaction is intuitively defined as how fast or slow a reaction takes place. For example, the oxidation of iron under the atmosphere is a slow reaction which can take many years.

First order reaction:

A reaction in which the first power of concentration term enters into the general equation is called first order reaction

Mathematical terms:

Mathematical Expression: Let us consider the first order reaction as follows :



Initially : $a \text{ mol L}^{-1}$

After time, t : $(a-x) \text{ mol L}^{-1}$

If the initial concentration of reactant A is 'a' mol and after time 't' x mol of A has been converted to product B then $(a - x)$ mol of A remains. So according to Law of mass action, rate of first order reaction at that time will be :

$$\frac{dx}{dt} = k \times (a - x)^1$$

or, $\frac{dx}{(a-x)} = k \times dt$, here k is a constant.

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On integrating both sides we get:

$$\int \frac{dx}{(a-x)} = \int (k \times dt),$$

or, $-\ln(a-x) = k \times t + c$ (integration constant)

When, $t = 0$, then $x = 0$, so $-\ln a = c$ (const). Putting this value in equation (1) we get.

$$-\ln(a-x) = k \times t - \ln a \quad \text{or } k \times t = \ln \frac{a}{a-x}$$

$$\text{Or, } k = \frac{1}{t} \times \ln \frac{a}{a-x}$$

$$\text{or, } k = \frac{2.303}{t} \times \ln \frac{a}{a-x} \quad \text{this is the rate constant of a first order reaction.}$$

(A) Unit of k of first order reaction : From the equation for rate constant k of a first

$$\text{order reaction: } k = \frac{2.303}{t} \times \ln \frac{a}{a-x}, \text{ we get unit of } k = s^{-1} \text{ or } \text{min}^{-1}$$

Characteristics of first order reaction :

There are three characteristics of first order reaction as follows

(i) A first order reaction never completes.

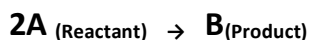
(ii) The time needed to complete a definite portion or 50% of the reaction does not

depend on initial concentration of the reactant.

(iii) The rate constant (k) of first order reaction does not depend on unit Of concentration of reactants.

- **2nd Order reaction:** - The reaction whose rate is proportional to the square of concentration of one reactant or to the concentrations of two reactants is called a second order reaction.
- **Rate equation for 2nd order reaction: let us consider the following 2nd order reaction,**

2A → products



Initially, $t = 0$, $a \text{ moleL}^{-1} \rightarrow 0$

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After time, $t = t$, $(a-x)$ mole $L^{-1} \rightarrow x$

If the initial concentration of A be a mol L^{-1} and if in time “ t ” the concentration of A will be $(a-x)$ mol/L. So according to law of mass action at any time t the rate of reaction will be:

$$\frac{dx}{dt} \propto (a-x)^2$$

$$\Rightarrow \frac{dx}{dt} = k (a-x)^2$$

$$\Rightarrow \frac{dx}{(a-x)^2} = k dt$$

$$\Rightarrow \int \frac{dx}{(a-x)^2} = \int k dt$$

$$\Rightarrow \frac{1}{a-x} = kt + C \dots \dots \dots (i)$$

When, $t=0$, $x=0$ then from eqⁿ (i) we get,

$$\frac{1}{a} = 0 + C$$

$$\Rightarrow C = \frac{1}{a} \dots \dots \dots (ii)$$

Putting the value of C in eqⁿ (i) we get,

$$\frac{1}{a-x} = kt + \frac{1}{a}$$

$$\Rightarrow kt + \frac{1}{a} = \frac{1}{a-x}$$

$$\Rightarrow Kt = \frac{1}{a-x} - \frac{1}{a}$$

$$\Rightarrow Kt = \frac{a-a+x}{a(a-x)}$$

$$\Rightarrow Kt = \frac{x}{a(a-x)}$$

$$\Rightarrow K = \frac{1}{t} \cdot \frac{x}{a(a-x)}$$

THE CHARACTERISTICS OF SECOND ORDER REACTIONS:

The following are the two main characteristics second order reaction:

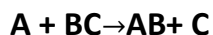
- 1) In a second order reaction involving one reactant, the time needed for completion of certain fraction or 50% of reactant is inversely proportional to initial concentration of the reactant.
- 2) A second order reaction involving two reactants becomes first order when one reactant is in larger excess than the other.

Activation Energy and Activated Complex -Transition State Theory

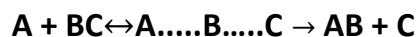
From the Arrhenius equation, it is evident that only those molecules can participate in reactions, which have at least some definite higher energy than the average energy of the molecule. Excess energy is known as activation energy.

This difference between the average energy of reactant molecules and the energy which a reactant molecule must possess to enable it to take part in a reaction is known as activation energy of the reaction and the molecules which possess at least this amount of energy are called activated molecules.

Only the activated molecules can take part in the reaction and other molecules cannot till they acquire necessary activation energy. When two activated molecules come in contact with each other, they form an activated complex. Then the activated complex decomposes to the products. For example, let us consider the reaction,



When A molecule approaches BC molecule, a bond between A and B gradually forms and the bond between B and C gradually becomes weak. At certain stage A, B and C form an activated complex, which then decomposes to AB and C.



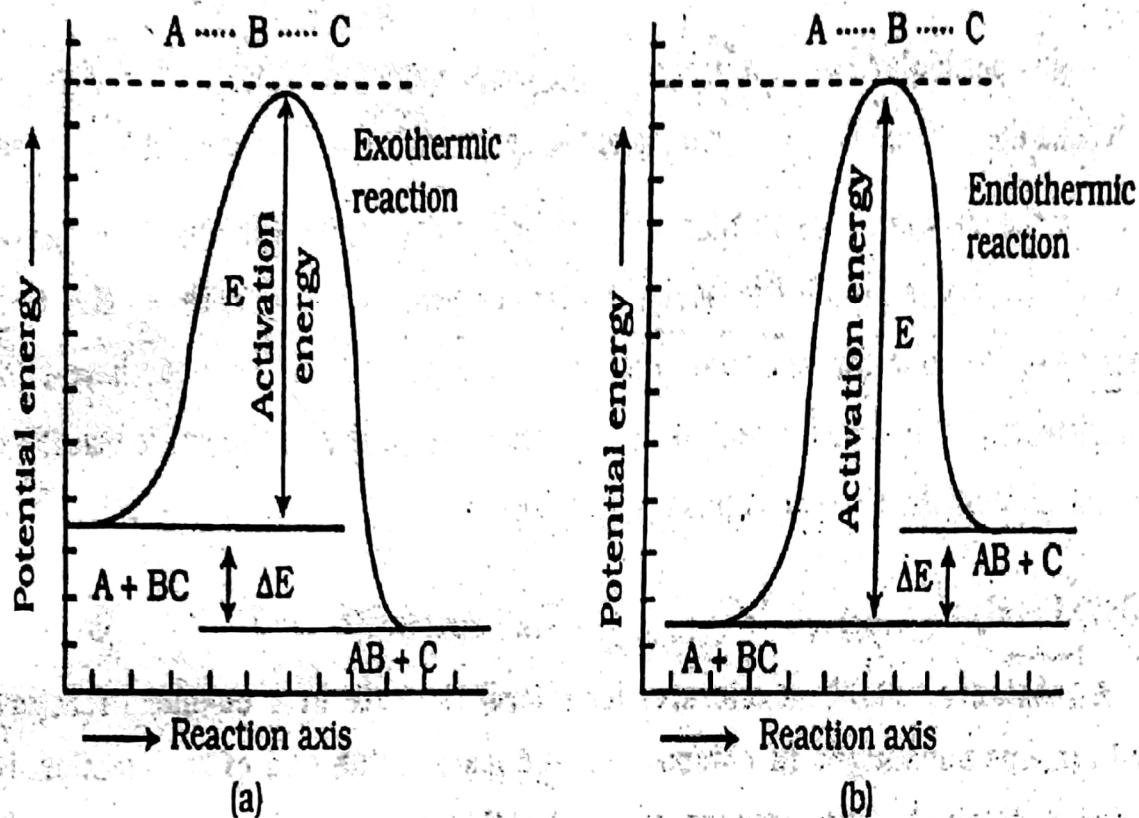


Fig. 13.10 : Potential energy versus reaction axis

The rate of the total reaction is the rate of formation of the activated complex. Hence the higher is the activation energy, the lower is the rate the reaction and vice versa. Because if the activation energy is high, then it becomes different for reactant molecules to acquire this amount of energy: hence the reaction becomes slow. On the other hand if the activation energy is low, then many more molecules can acquire this energy and can participate in the reaction Hence the rate of the reaction becomes high.

Collision Theory of Chemical Reaction

Collision theory was successfully applied to explain the bimolecular reactions in gaseous phase, but later it was found to be applicable also to the reaction in solution.

For reaction to take place :

- The first condition is- The Particles (atom, ion or molecules) taking part in a reaction must collide with one another.
- The second condition is- The collisions must take place with a minimum amount of energy.
- The third condition is- The collisions must take place with definite orientation. Arrhenius equation is in fact the mathematical expression of these conditions.

From the Arrhenius equation :

$$k = p.Z.e^{-\frac{E_a}{RT}}$$

where,

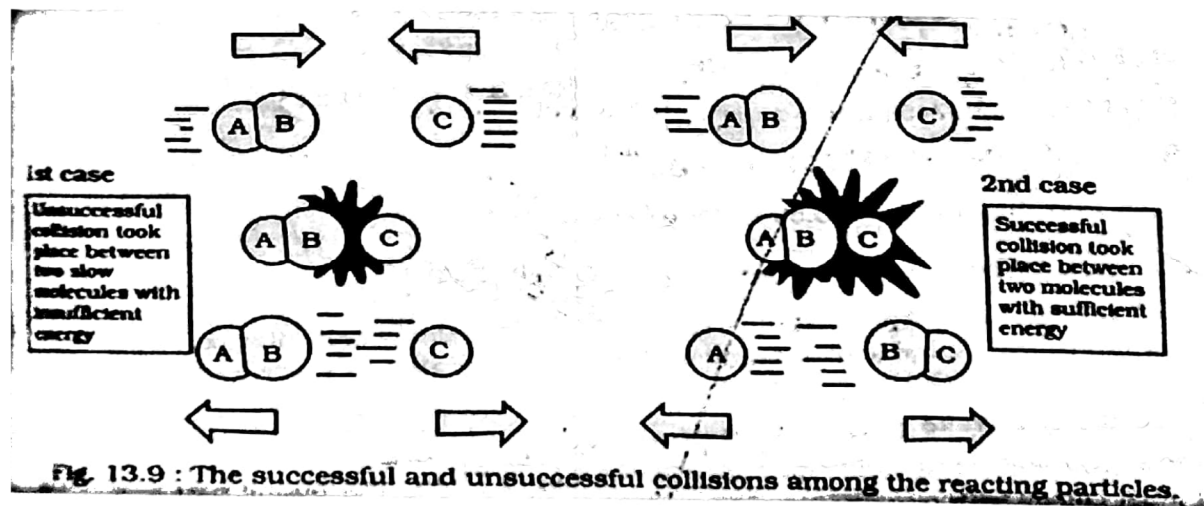
k = rate constant of reaction ;

p = steric factor (condition no. 3) ;

z = collision rate (condition no. 1) ;

$e^{-\frac{E_a}{RT}}$ = fraction of molecules with activation energy (condition no. 2) .

Explanation: According to the first and second conditions the reacting particles must collide with one another with minimum energy. Then some of the old chemical bonds will be broken down and new bonds will be formed. The collisions may be shown as in following figure.



From the figure it is evident that in the first case the collision took place with insufficient energy. From and hence the breaking of old bonds did not take place. The molecule state. But in the second case the collision took place with sufficient energy between A and B broke down and a new bond between B and C was formed. Molecule BC is formed.

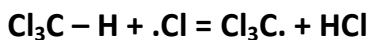
According to the third condition, the collision has to take place in definite orientation. This may be explained as follows :

The attacking reactant must come in contact with the particular atom, with which bond breaking and formation will take place, in a suitable orientation. If this atom is attached to a large group and the attacking reactant comes from that side, then it will not be able to come into contact with that atom. **The hindrance of large group to the approaching reactant is known as "steric hindrance".**

Besides this hindrance for breaking of old bond and formation of bond, collision should take place in definite orientation. **For example**, chloroform molecule: one carbon atom is bonded tetrahedrally to three chlorine atoms and one hydrogen atom. During the photochemical reaction between CHCl_3 and CHCl_2 , chlorine molecule is first dissociated into two active chlorine atoms.



Now if the chlorine atom approaches the chloroform molecule from the side of Hydrogen atom, then the reaction takes place.



But if the chlorine atom approaches the chloroform molecule from the side of chlorine then no reaction will take place.

